DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination 2023

Course: B. Tech. Branch: Electronics & Telecommunication Engg./Electronics Engg Semester :VI

Subject Code & Name: BTETPE603E/BTEXPE603A Information Theory and Coding

Max Marks: 60 Date: 17/07/2023 Duration: 3 Hr.

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		
A)	In a factory, four machines A_1, A_2, A_3 and A_4 produce 10%, 20%, 30% and 40% of the items, respectively. The percentage of defective items produced by them is 5%, 4%, 3% and 2%, respectively. An item selected at random is found to be defective. What is the probability that it was produced by the machine A_2?	**L1**	**6**
B)	Define CDF. State the properties of CDF.	**L1**	**6**
C)	Write a note on Ergodic process	**L1**	**6**
Q.2	**Solve Any Two of the following.**		
A)	An urn contains 4 white and 3 black balls. Two balls are drawn successively with X denoting the number of black balls. a. Find the probability function of X b. Draw the bar chart and histogram.	**L2**	**6**
B)	Write a note on power spectral density and prove its properties.	**L1**	**6**
C)	Explain the Resistor noise with the help of an equivalent diagram and power density spectrum.	**L1**	**6**
Q. 3	**Solve Any Two of the following.**		
A)	Write an note on joint and conditional entropy.	**L3**	**6**
B)	What is mutual information? Derive its expression from entropies.	**L2**	**6**
C)	Also the Shannon-Fano coding procedure for the following message ensemble: $[X] = [\ x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7]$ $[Y] = [0.4 \quad 0.2 \quad 0.12 \quad 0.08 \quad 0.08 \quad 0.08 \quad 0.04]$ Take M=2.	**L2**	**6**

Q.4	**Solve Any Two of the following.**		
A)	For a systematic linear block code, the three parity check digits are given by $C4 = d_1 + d_2 + d_3$ $C5 = d_1 + d_2$ $C6 = d_1 + d_3$ i) Construct generator matrix ii) Construct code generated by this matrix iii) Determine error correcting capability iv) Prepare a suitable decoding table v) Decode the received code words 101100 and 000110	**L3**	**6**
B)	Write a note on convolution encoder. Draw its basic structure and explain.	**L2**	**6**
C)	Write a note on Syndrome decoding. What happens if double error occurs in received code word when the minimum distance criterion is 3?	**L3**	**6**
Q. 5	**Solve Any Two of the following.**		
A)	Discuss any three types of vocoders in brief.	**L1**	**6**
B)	Write a short note on vector quantization.	**L2**	**6**
C)	Explain GSM codec.	**L2**	**6**